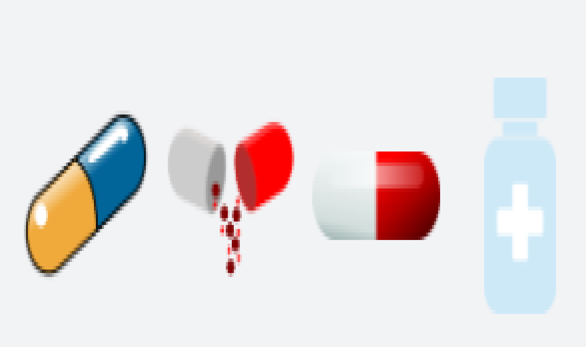
University of Al-Kafeel
College of Dentistry



Lec-6

Adrenergic antagonist Drugs

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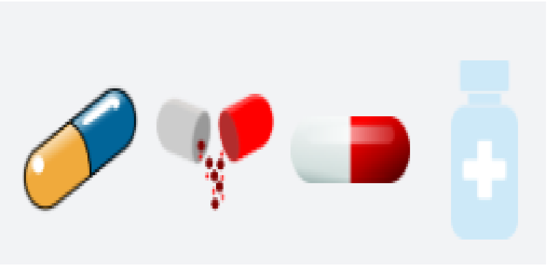
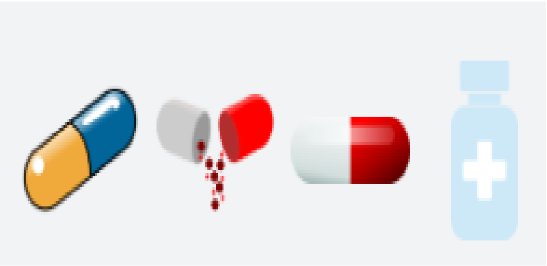
Learning Objectives



By the end of this lecture, you should be able to

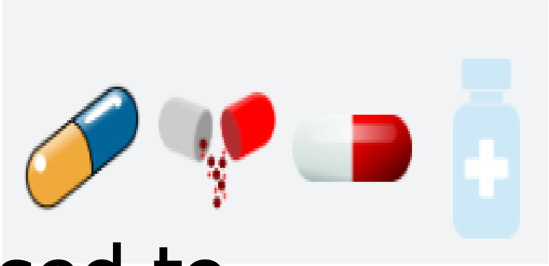
- Explain the mechanism of action of adrenergic antagonists .
- Identify the therapeutic uses and clinical indications of adrenergic antagonists
- Recognize the main adverse effects, contraindications, and precautions associated with adrenergic antagonist drugs.

ADRENERGIC ANTAGONISTS



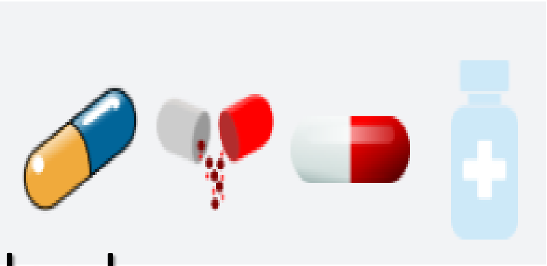
- The adrenergic antagonists (also called blockers or **sympatholytic** agents) bind to adrenoceptors .
- The adrenergic antagonists are classified according to their relative affinities for α or β receptors in the peripheral nervous system.
- These drugs will interfere with the functions of the sympathetic nervous system.

Alpha-adrenergic antagonists



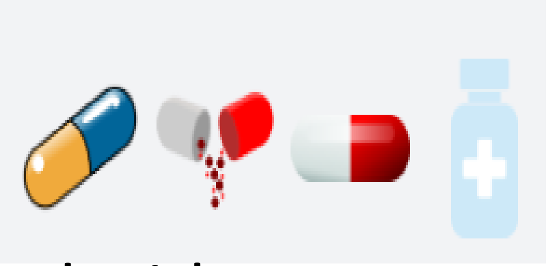
- also known as alpha-blockers, are primarily used to treat **hypertension** and **benign prostatic hyperplasia (BPH)**.
- They work by blocking alpha-1 receptors to relax smooth muscles in blood vessels, lowering blood pressure, and in the prostate and bladder neck, which improves urine flow.
- Other uses can include managing urinary stones, post-traumatic stress disorder, and managing symptoms associated with certain medical emergencies, such as overdose from stimulants.

Therapeutic uses of alpha-adrenergic antagonists



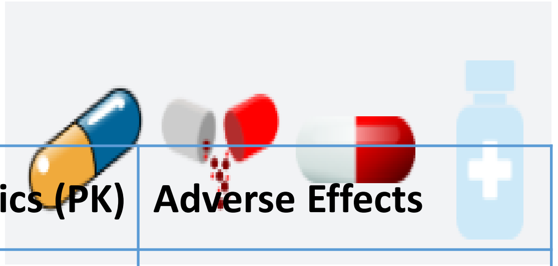
- Hypertension: By relaxing blood vessel muscles, these drugs lower blood pressure. However, they are generally recommended as an adjunctive therapy, not a first-line monotherapy, due to potential side effects like postural hypotension.
- Benign Prostatic Hyperplasia (BPH): Alpha-blockers relax the smooth muscles in the prostate and bladder neck, easing symptoms of urinary obstruction such as a weak stream and difficulty urinating.
- Pheochromocytoma: In patients with this adrenal gland tumor, which releases excessive amounts of catecholamines, alpha-blockers are used to manage the resulting high blood pressure. They are used as adjunctive therapy alongside beta-blockers.

Other uses



- Urinary stones: Can be used to help relieve pain associated with ureteric stones.
- Post-traumatic stress disorder (PTSD): Certain alpha-blockers are used in managing PTSD symptoms.
- Cardiovascular toxicity: Alpha-blockers can be used in emergency situations to treat acute toxicity from stimulant overdoses, like those from amphetamines or cocaine.
- Anxiety and panic disorders: Some uses have been explored for these conditions.

α- adrenergic antagonists



Drug	Receptor Selectivity	Action	Clinical Uses	Pharmacokinetics (PK)	Adverse Effects
Phenoxybenzamine	Nonselective α_1/α_2 (irreversible)	Long-lasting vasodilation	Pheochromocytoma	long duration (24–48 h) due to covalent receptor binding; hepatic metabolism	Postural hypotension, reflex tachycardia, nasal congestion
Phentolamine	Nonselective α_1/α_2 (competitive)	Vasodilation; ↑ NE release	Hypertensive crisis, pheochromocytoma	Rapid onset (IV), short half-life (≈19 min), renal excretion	Tachycardia, arrhythmias, hypotension
Prazosin, Terazosin Doxazosin	Selective α_1	Vasodilation; ↓ peripheral resistance	Hypertension, BPH	oral absorption; hepatic metabolism; high first-pass effect	Postural hypotension
Tamsulosin	α_1-selective	Relaxation of prostate/bladder	BPH	Half-life: 9–13 h; hepatic metabolism	Ejaculatory dysfunction, dizziness
Yohimbine	Selective α_2 antagonist	↑ NE release	Rare use	Rapid absorption; half-life ≈0.6–1 h; hepatic metabolism	Anxiety, hypertension, tachycardia

α- ADRENERGIC ANTAGONISTS

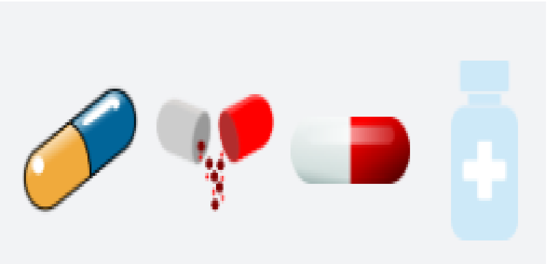
Blocks α-adrenergic receptors to modify physiological responses.

DRUG (Icons of Pills/Molecules)	RECEPTOR SELECTIVITY (Diagrams of Receptors)	ACTION (Arrows & Physiology Icons)	CLINICAL USES (Medical Condition Icons)	ADVERSE EFFECTS (Warning Symbols & Symptoms)
PHENOXYBENZAMINE	α1 α2 Nonselective α1/α2 (Irreversible/Competitive)	Long-lasting vasodilation ↑ NE release	Pheochromocytoma	Postural hypotension, reflex tachycardia, nasal congestion
PHENTOLAMINE			Hypertensive crisis, pheochromocytoma	Tachycardia, arrhythmias, hypotension
PRAZOSIN, TERAZOSIN, DOXAZOSIN	α1 Selective α1	Vasodilation; ↓ peripheral resistance Relaxation of prostate/bladder	Hypertension, BPH	Postural hypotension
TAMSULOSIN			BPH	Ejaculatory dysfunction, dizziness
YOHIMBINE	α2 Selective α2 antagonist	↑ NE release	? Rare use	Anxiety, hypertension, tachycardia

β-ADRENERGIC BLOCKING AGENTS

- Selective β blockers act at both β1 and β2 receptors, whereas cardio-selective β antagonists primarily block β1 receptors
- B-blockers are also effective in treating angina, cardiac arrhythmias, myocardial infarction, congestive heart failure, hyperthyroidism, and glaucoma as well as serving in the prophylaxis of migraine headaches.
- The names of all β-blockers end in “-olol” except for *labetalol* and *carvedilol*.

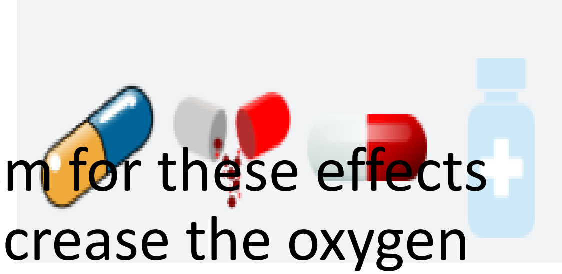
Therapeutic uses of β -blockers



- **a. Hypertension**
- β -blockers does not reduce blood pressure in people with normal blood pressure.
- β -blockers lowers blood pressure in hypertension by several different mechanisms of action. Decreased cardiac output is the primary mechanism, inhibition of renin release from the kidney, decrease in total peripheral resistance with long-term use, and decreased sympathetic outflow from the CNS also contribute to the antihypertensive effects.

  b. Angina pectoris
- β -blockers decreases the oxygen requirement of heart muscle and, therefore, is effective in reducing chest pain on exertion that is common in angina. Propranolol is therefore useful in the management of chronic stable angina.

Therapeutic uses



- **c. Myocardial infarction**
- β -blockers have a protective effect on the myocardium. The mechanism for these effects may be a reduction in the actions of circulating catecholamines that increase the oxygen demand in an already ischemic heart muscle. *Propranolol* also reduces the incidence of sudden arrhythmic death after myocardial infarction.
- **d. Migraine**
- β -blockers *is effective in reducing migraine episodes when used prophylactically. It is one of the more useful β -blockers for this indication, due to its lipophilic nature that allows it to penetrate the CNS.*
- **e. Hyperthyroidism**
- *β -blockers are effective in blunting the widespread sympathetic stimulation that occurs in hyperthyroidism. In acute hyperthyroidism (thyroid storm), β -blockers may be lifesaving in protecting against serious cardiac arrhythmias.*

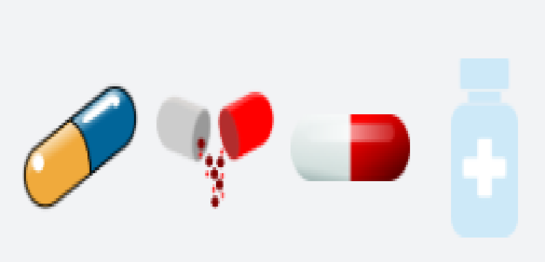
β-Blockers

Group / Action	Drugs	Mechanism of action	Clinical Uses	Adverse Effects
β_1-Selective (Cardioselective)	Metoprolol, Atenolol, Bisoprolol, Esmolol, Nebivolol, Acebutolol	Block β_1 receptors → ↓ HR, ↓ contractility, ↓ renin	HTN, angina, arrhythmias, post-MI, HF (metoprolol, bisoprolol),	Bradycardia, AV block, fatigue, masks hypoglycemia.
Nonselective $\beta_1 + \beta_2$ Blockers	Propranolol, Nadolol, Timolol, Sotalol	Block β_1 & β_2 → ↓ HR + bronchoconstriction & ↓ glycogenolysis	HTN, angina, essential tremor, hyperthyroidism, migraine prophylaxis (propranolol), glaucoma (timolol), arrhythmias (sotalol)	Bronchospasm, bradycardia, fatigue, depression (propranolol), worsened hypoglycemia
β-Blockers With ISA (Partial Agonists)	Pindolol, Acebutolol, Carteolol	Partial β-agonist + antagonist → less resting bradycardia	Hypertension (less commonly used), patients with bradycardia who still need β-blocker	Mild tachycardia possible, less bradycardia, not preferred post-MI
Mixed α_1 + β Blockers	Labetalol, Carvedilol	Block $\beta_1/\beta_2 + \alpha_1$ → ↓ HR + vasodilation	Hypertensive emergencies (labetalol), pregnancy HTN, heart failure (carvedilol)	Postural hypotension, dizziness, bradycardia
β-Blockers With Additional Vasodilation	Nebivolol, Carvedilol	Nebivolol: β_1 block + ↑ NO release; Carvedilol: β block + α_1 block	Hypertension, HF (carvedilol)	Hypotension, bradycardia

BETA-BLOCKERS (β-BLOCKERS)

Class	Drugs	Mechanism	Clinical Uses	Adverse Effects
β_1-Selective (Cardioselective)	Metoprolol, Atenolol Bisoprolol, Esmolol Nebivolol, Acbivolol Acebutolol	Block β1 receptors ↓ HR ↓ contractility ↓ renin	HF HTN, angina, arrhythmias, post-MI, HF (metoprolol, bisoprolol)	AV Bradycardia, AV block, fatigue, masks hypoglycemia; less bronchospasm
Nonselective β_1 + β_2 Blockers	Propranolol Nadolol Timolol Sotalol	Block β1 & β2 ↓ HR + bronchoconstriction ↓ glycogenolysis	HTN, angina, essential tremor, hyperthyroidism, migraine prophylaxis (propranolol), glaucoma (timolol), arrhythmias (sotalol)	Bronchospasm, bradycardia, fatigue, depression (propranolol), worsened hypoglycemia
β-Blockers With ISA (Partial Agonists)	Pindolol Acebutolol Carteolol	Partial β-agonist + antagonist, less resting bradycardia	Hypertension (less used), patients with bradycardia needing β-blocker	Mild tachycardia possible, less bradycardia, not preferred post-MI
Mixed α1 + β Blockers	Labetalol Carvedilol	Block β1/β2 + α1 ↓ HR + vasodilation	Hypertensive emergencies (labetalol), pregnancy HTN, heart failure (carvedilol)	Postural hypotension, dizziness, bradycardia
β-Blockers With Additional Vasodilation	Nebivolol Carvedilol	Nebivolol: β1 block + ↑ NO release Carvedilol: β block + α1 block	Hypertension, HF (carvedilol)	Hypotension, bradycardia

References



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THANK
YOU!